Web Design & Development

Detailed Chapter Notes & Worked Examples - Volume 2

This pack develops practical understanding through explanations, worked examples, lab tasks and review questions. Complete the matching Foundation Pack first.

Chapter	Topic
1	Semantic HTML
2	Responsive CSS
3	JavaScript interaction
4	Deployment and quality

Chapter 1: Semantic HTML

Concept notes

Semantic elements describe meaning and improve accessibility and maintenance. Labels connect to form controls; alt text explains meaningful images.

Worked example

Use header, nav, main, section and footer; each form input has label and appropriate type.

Lab assignment

Build an accessible course enquiry page and validate structure.

Review questions

1. Why semantic HTML?
2. What makes good alt text?
3. Why label form controls?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 2: Responsive CSS

Concept notes

Mobile-first CSS starts with small screens, then enhances through media queries. Flexbox handles one-dimensional layout; Grid handles rows and columns.

Worked example

Cards use grid-template-columns: repeat(auto-fit,minmax(240px,1fr)); images use max-width:100%.

Lab assignment

Build responsive course cards and test 360, 768 and 1366 widths.

Review questions

1. When use Grid?
2. What causes overflow?
3. Why mobile-first?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 3: JavaScript interaction

Concept notes

Events trigger functions; DOM methods read or update the page. Validate for usability but repeat validation on server for security.

Worked example

On submit, trim name, test phone pattern, show errors beside fields and focus first invalid field.

Lab assignment

Add accessible menu and client-side enquiry checks.

Review questions

1. Why server validation too?
2. What is an event listener?
3. How show accessible errors?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Chapter 4: Deployment and quality

Concept notes

Production requires correct URL, HTTPS, environment settings, backups, caching and testing. Never expose secrets in repository or browser code.

Worked example

Deploy from Git, keep .env private, clear/cache framework settings and test key routes and mobile layout.

Lab assignment

Publish a multi-page site with deployment checklist and rollback note.

Review questions

1. Why keep .env private?
2. What is rollback?
3. Which routes need smoke testing?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

Final Applied Assignment

Create one original project that combines all four chapters of Web Design & Development. Use realistic but non-sensitive sample data. Submit planning notes, source files, final output, test evidence and a one-page reflection.

Assessment rubric

Area	Marks
Concept accuracy	25
Working practical output	30
Validation and testing	20
File organisation and presentation	15
Original explanation	10

Student declaration

I confirm that the submitted work is my own practice. I have not included passwords, OTPs, identity documents or confidential student records.

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